TITLE PROPOSALS

ITSAND 313: Systems Analysis and Design

GROUP NAME: Trio-I & Error

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YR & SECTION: 3-1

TITLE 1:	**Web-Based Sales, Inventory, Customer, and Location Mapping Management System for a Water Refilling Station**
DESCRIPTION:	This system is designed for a single water refilling station to manage its daily business operations through a centralized web-based platform. It covers sales transactions, product inventory, customer records, orders, payments, deliveries, and reporting. Staff can record orders and purchases, monitor product availability, manage customer information, process payments, and track deliveries, while the administrator can monitor sales performance, inventory levels, and generate reports for business analysis and decision-making. The system also supports vending machines owned by the station; sales made through each machine are recorded and automatically added to the station's daily and overall sales total. In addition, a Location Mapping and Geolocation feature allows the station to record, visualize, and manage customer delivery locations through an interactive digital map, helping staff and delivery personnel identify delivery destinations and plan deliveries more efficiently. The system aims to reduce manual record-keeping, minimize errors in sales and inventory tracking, improve customer and delivery management, consolidate sales from the station and its vending machines, and provide an organized and efficient way of managing the daily operations of a water refilling station.
FUNCTIONALITIES AND FEATURES:	**CORE FUNCTIONALITIES:** ● **User Registration and Login** – Allows the administrator and staff to securely access the system according to their assigned roles and permissions. ● **Staff Account Management** – Allows the administrator to create, update, deactivate, and manage staff accounts and access permissions. ● **Customer Registration** – Allows staff to record customer information such as name, contact number, address, customer type, and delivery location. ● **Customer Profile Management** – Allows staff to view and update customer information, including contact details, delivery addresses,

	and transaction history. • **Product Management** – Allows the administrator to manage products such as purified water, mineral water, water containers, caps, bottles, and other station supplies. • **Product Pricing Management** – Allows the administrator to set, update, and manage product prices used in sales transactions. • **Sales Transaction Management** – Allows staff to record customer purchases, calculate the total amount, and maintain sales transaction records. • **Vending Machine Sales Recording** – Allows staff to record sales from each vending machine owned by the station and automatically adds them to the daily and overall sales total. • **Order Management** – Allows staff to record customer orders for pickup or delivery, including order details, quantities, customer information, and requested schedules. • **Container Management** – Tracks available, borrowed, returned, damaged, and customer-owned water containers. • **Inventory Management** – Monitors the quantity and availability of water containers, caps, bottles, and other station supplies. • **Stock-In Management** – Records newly purchased or added products and supplies and updates the corresponding inventory quantities. • **Stock-Out Management** – Records products and supplies released, sold, consumed, or damaged during business operations and updates inventory quantities. • **Delivery Management** – Records customer delivery requests, delivery schedules, assigned personnel, delivery destinations, and delivery status. • **Payment Recording** – Records payment amounts, payment methods, and payment status. • **Location Mapping and Geolocation** – Allows staff to record, locate, and visualize customer delivery addresses and the water refilling station using an interactive digital map. • **Customer Location Management** – Allows staff to add, update, and manage customer delivery locations, including map coordinates and multiple delivery addresses when necessary. • **Reports Generation** – Generates sales, inventory, customer, product, payment, order, container, delivery, vending machine, and location-based reports.

	SYSTEM FEATURES: ● **Dashboard** – Displays important information such as total sales (including vending machine sales), pending orders, completed deliveries, available containers, low-stock products, and customer or delivery locations. ● **Daily Sales Monitoring** – Allows the administrator to monitor daily sales from the station and vending machines, and completed transactions. ● **Sales Summary** – Provides summaries of daily, weekly, and monthly sales, with vending machine sales included in the totals. ● **Low-Stock Monitoring** – Identifies products and supplies that have reached their minimum stock level and provides alerts for items that need replenishment, including vending machine stock levels. ● **Delivery Status Tracking** – Tracks orders as Pending, Preparing, Out for Delivery, Delivered, or Cancelled. ● **Delivery Area Monitoring** – Displays customer and delivery locations on a map to help the administrator monitor service areas and identify areas with frequent deliveries. ● **Delivery Location Reference** – Provides delivery personnel with a visual map reference of customer destinations to help them locate delivery addresses more accurately. ● **Customer Transaction History** – Allows staff to view previous purchases, orders, payments, and deliveries of customers. ● **Search and Filtering** – Allows staff to search customers, products, orders, transactions, and delivery records using relevant information such as names, dates, status, and locations. ● **Printable Transaction Records** – Allows staff to print receipts, order records, payment records, or other transaction records when needed. ● **Activity Logs** – Records important actions performed by the administrator and staff for monitoring, accountability, and security purposes. ● **Account Management** – Allows users to update their account information and password.
PLATFORM:	Web-Based Platform